SILAS KWABLA GAH

PhD Candidate in Computer Vision and Machine Learning · University of Ghana

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To:
Prof. Dr. Djamila Aouada
Head of Computer Vision, Machine Intelligence and Imaging (CVI^2)
Interdisciplinary Centre for Security, Reliability and Trust (SnT)
University of Luxembourg
29, Avenue John F. Kennedy, L-1855 Luxembourg

Dear Prof. Aouada and Members of the Selection Committee,

I am writing to express my enthusiastic application for the **Postdoctoral Researcher in Computer Vision** position within the Computer Vision, Machine Intelligence and Imaging (CVI^2) research group at the Interdisciplinary Centre for Security, Reliability and Trust (SnT), University of Luxembourg. As a final-year PhD candidate in Computer Vision and Machine Learning at the University of Ghana (thesis defence scheduled for October 2026), my doctoral research investigates **reliable 3D geometric perception**, **inference-time foundation-model coordination ($\Delta\theta = 0$)**, **probabilistic calibration under distribution shift**, and **self-supervised multi-view consensus**. My background directly aligns with CVI^2's pioneering research in 3D scene understanding, multi-sensor perception, and trustworthy visual AI.

Alignment with SnT's Mission: Reliability, Trust, and Calibrated Perception

The defining mission of SnT—establishing foundational **security, reliability, and trust** in autonomous systems—is the central theme of my doctoral dissertation, *Visual Perception under Low Supervision: Inference-Time Coordination of Frozen Foundation Model Representations.* In real-world environments, deep perception models frequently produce overconfident yet erroneous predictions when encountering novel semantic categories or sensor variations.

To overcome this fundamental vulnerability without requiring costly task-specific retraining, I formulated **Reliability-Calibrated Adaptive Semantic Arbitration (RCASA)** (currently in preparation for *IEEE TPAMI*). RCASA provides a formal probabilistic framework that explicitly separates an encoder's internal candidate confidence from its contextual reliability via Platt scaling and Bernoulli uncertainty estimation. By proving class-wise boundedness and probability simplex preservation, the framework guarantees that multi-model decisions remain mathematically calibrated and robust against out-of-distribution hallucinations. At SnT, I am eager to expand these calibration principles to safety-critical perception, multi-sensor fusion, and autonomous operations.

3D Point Clouds, Geometric Lifting, and Overcoming Semantic Collapse

CVI^2's internationally recognized leadership in **3D shape analysis, geometric point clouds, and multi-sensor perception** strongly mirrors my experimental methodology:

- **3D Point-Cloud Segmentation on ScanNet200 (ACCV 2026, Paper #482):** I constructed training-free 3D geometric lifting pipelines that project camera-pose rays (K, T) to reconcile dense 3D vision-language representations (RegionPLC) with 2D promptable segmenters (SAM 2.1) through multi-view spatial consensus, establishing SOTA harmonic mean on challenging generalized few-shot indoor benchmarks.
- **Preventing Premature Semantic Collapse (ICLR 2027 Submission):** In our recent submission to **ICLR 2027**, titled *Beyond Argmax: A Mechanistic Study of Semantic Retention in Frozen Foundation-Model Composition for Generalized Few-Shot 3D Segmentation*, I identified

hard argmax decision collapse as a primary failure mode in multi-model pipelines. Across 312 3D scenes ($\sim$50M points), I proved that preserving compact top-k candidate distributions allows downstream geometric consensus to recover discarded novel classes, boosting segmentation harmonic mean from 28.47 to 34.87 (+6.40 points).

These methodologies directly support CVI^2's research portfolio, including high-precision 3D reconstruction, multi-camera sensor fusion, and spatial perception for proximity operations conducted in your state-of-the-art **Zero-G Lab**.

Physical Multi-View Consistency as Intrinsic Self-Supervision

In recent work submitted to SACAIR 2026, I formulated cross-view geometric consistency as an intrinsic, label-free process reward for multimodal reasoning. Developing an online **GRPO/VERL** reinforcement learning pipeline for **Qwen2.5-VL-7B** on **vLLM**, we converted multi-view physical agreement into a dense supervisory reward. Through counterfactual stress-testing across 1,000+ rollouts, we successfully neutralized reward-hacking failure modes (reducing spatial area bias correlation from $\rho > 0.72$ to 0.334) without requiring human reward labels. This experience bridges 3D geometric vision with generative multimodal foundation models, aligning with CVI^2's focus on visual generative modeling and self-supervised learning.

High-Performance Systems Rigor and Collaborative Independence

I bring rigorous systems engineering expertise in **PyTorch**, **CUDA/Triton** fundamentals, and large-scale evaluation pipelines. My implementations leverage float16 caching, chunked point processing, and streaming architectures to execute dense per-point evaluations efficiently on memory-bounded GPU clusters. This directly aligns with SnT's high-performance computing environment and the LuxProvide MeluXina supercomputer.

Beyond technical execution, having founded a successful software consultancy (EED Soft Consult) and lectured at Accra Technical University, I possess proven technical autonomy, project leadership, and mentorship capabilities. I have mentored undergraduate researchers and contributed to international peer review (SACAIR 2026 Program Committee).

I would be thrilled to join Prof. Aouada's group at SnT, contribute immediately to ongoing academic and industrial projects in 3D computer vision, and co-author high-impact publications at top-tier venues (CVPR, ICCV, ECCV, ICLR, TPAMI). Thank you very much for your time, consideration, and review of my application.

Sincerely,

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